Cyber Forensics

"OS-Forensics Practical Guide: Hands-On Cyber Forensics Toolkit"



OS Forensics v9,

A powerful suite used for digital investigations including disk imaging, system scanning, file indexing, signature verification, and memory analysis. This guide demonstrates practical steps and tool usage for real-world forensic scenarios.

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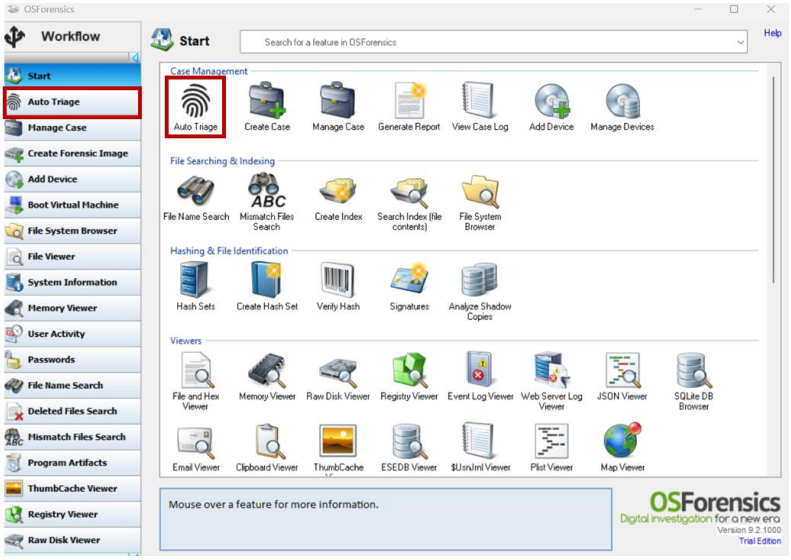
Case Report Generation **25**

OS Forensics



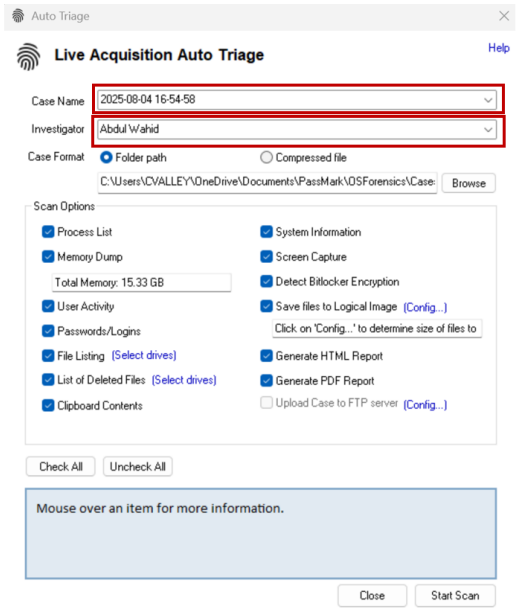
Auto Triage

- Open **OS-Forensics**.
- Select **Auto Triage** from the feature panel.
- This feature performs a quick scan of the system to gather important information like running processes, installed programs, system details, and recent activity.
- Useful for initial assessment and live response.



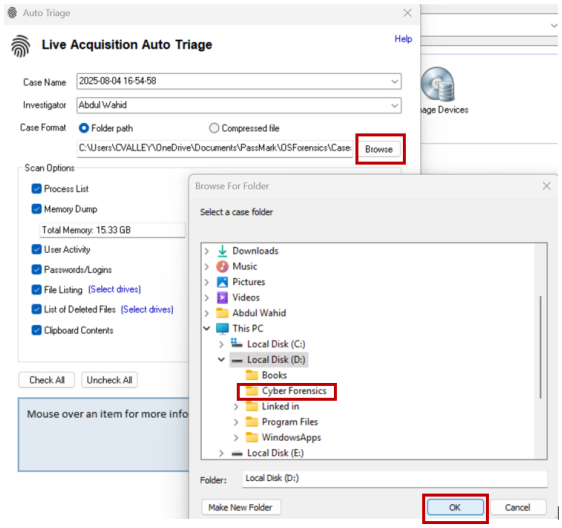
Fill in the required details:

- Case Name
- Investigator Name
- Case format

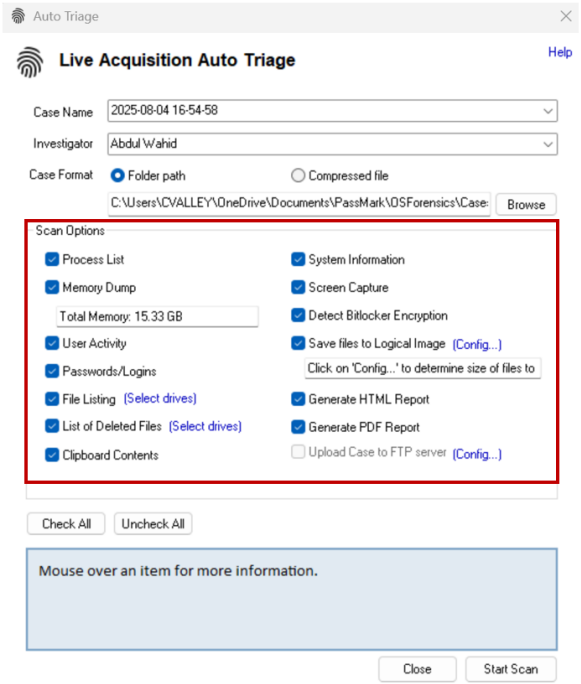


"Browse" the path but remember the directory should be Empty

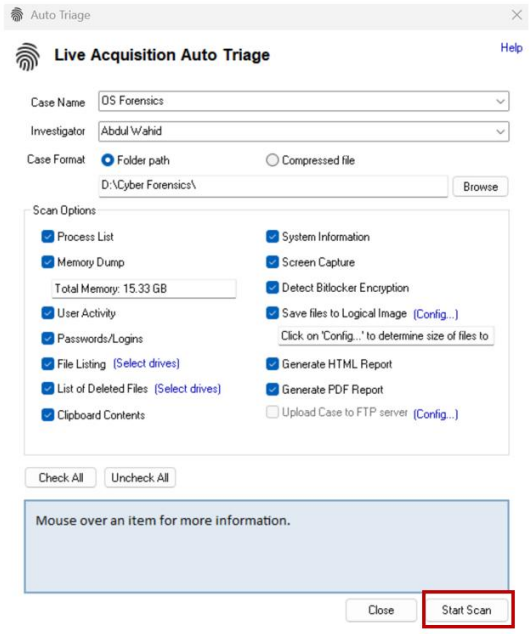
Select and then Click on "**ok**"



Now **check the options** on which you to perform scanning

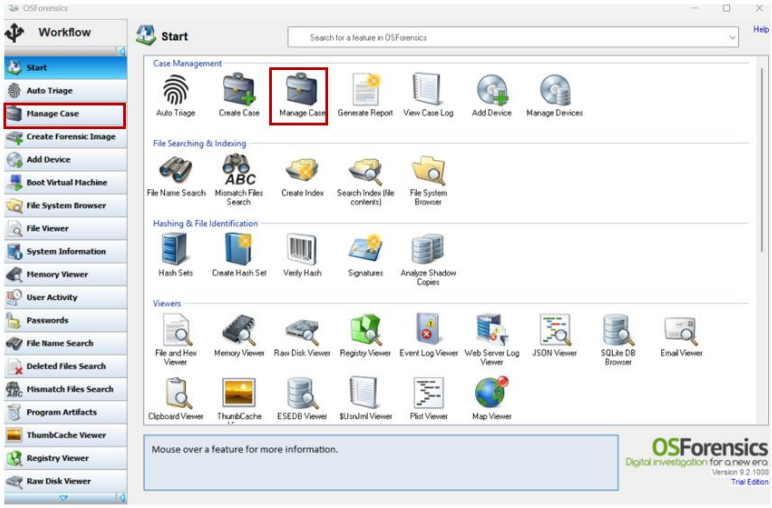


Click "**Start Scan**" to Perform Scanning



Managing Cases

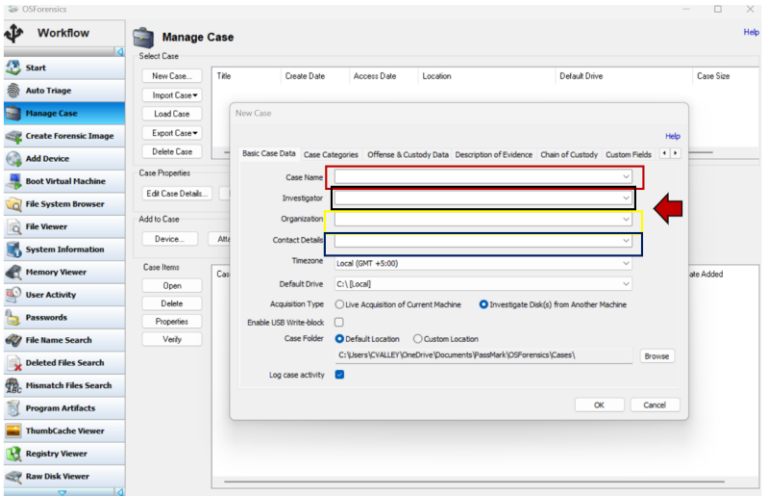
Select **Case Management** from the main menu.



Click on **New Case**.

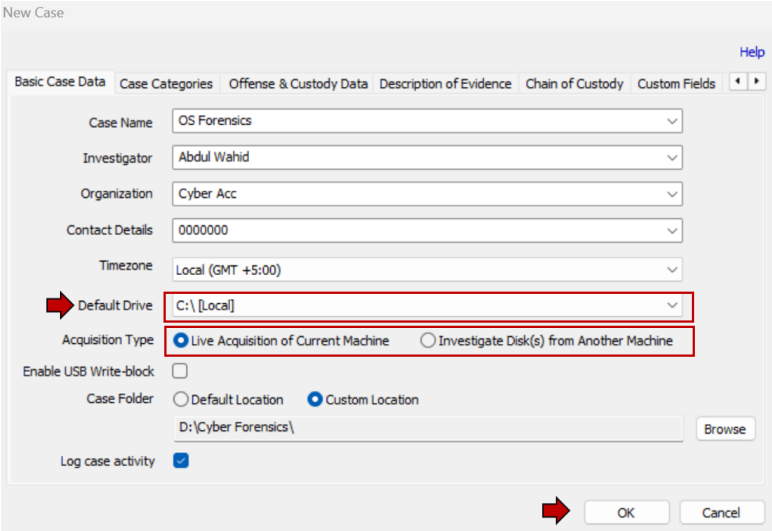
Fill in the required details:

- Case Name
- Investigator Name
- Organization
- Contact Details



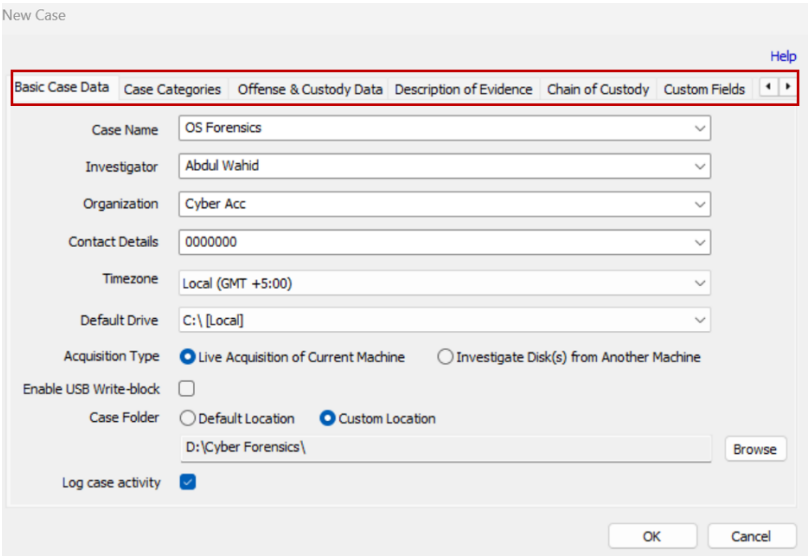
Choose one of the following:

- Live acquisition of the current machine
- Acquisition of another disk (e.g., from a connected hard drive or USB)
- Browse and select a folder (it must be empty) to store the case.
- Click **OK** to confirm.

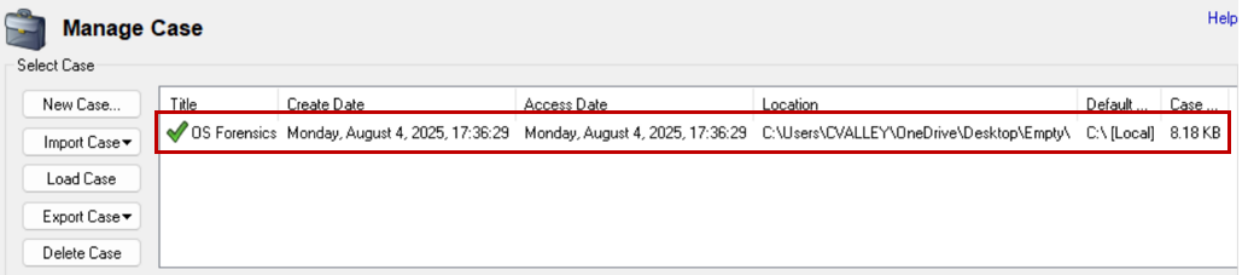


Enter additional details such as:

- Chain of custody
- Description of evidence

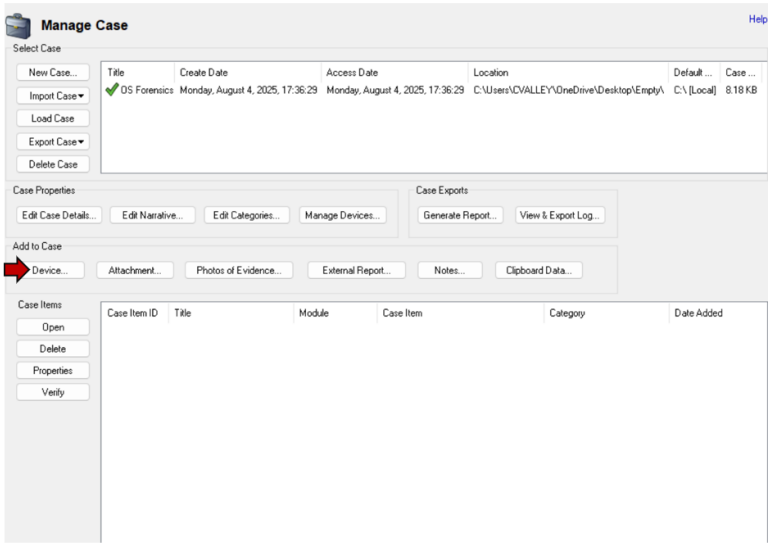


- A new case will now be created at your selected location.

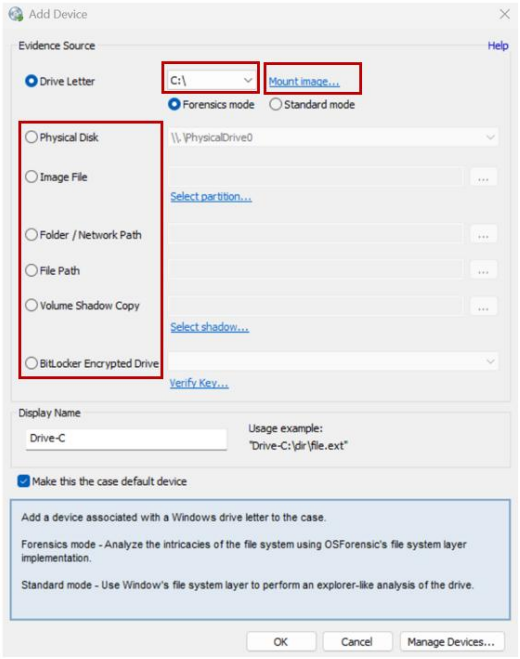


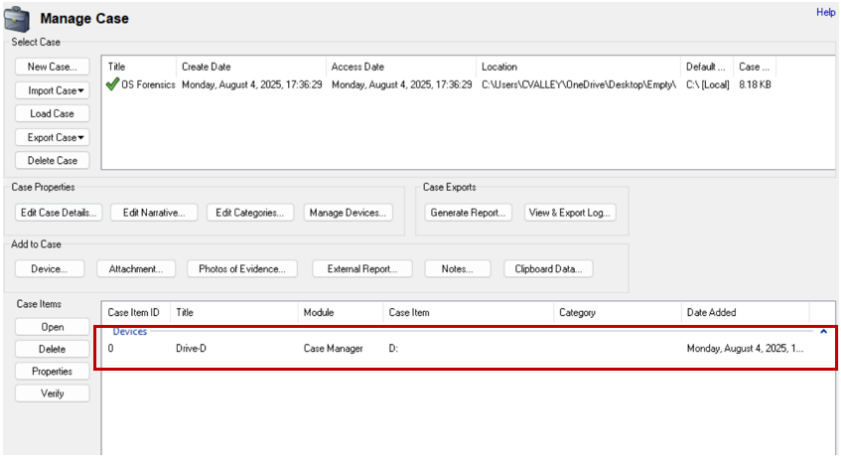
Adding a Device to the Case

- You can add evidence like a disk image, USB drive, or physical disk.
- Click on **Device** in the left panel.



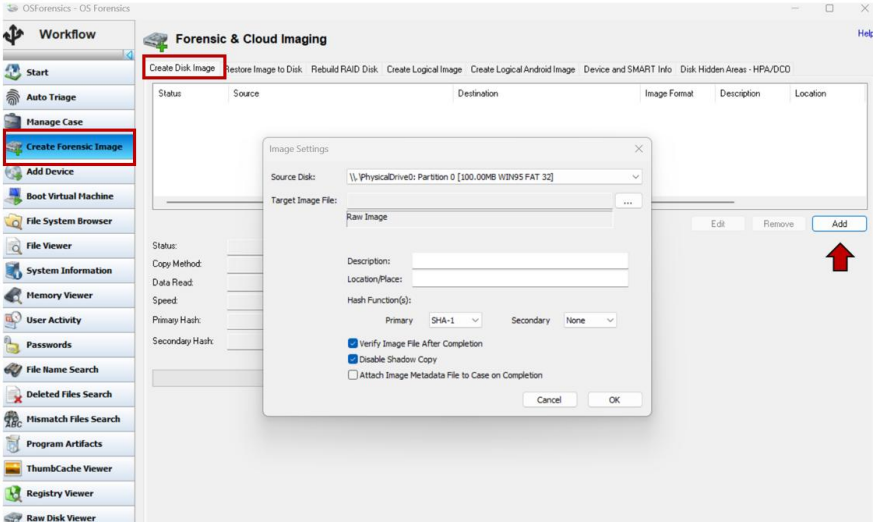
- To analyze an image file, click **Mount an Image**.
- This creates a **virtual drive** from your image file (e.g., E01, ISO) so you can browse it like a real drive.
- You can also analyze any physical or logical drive (e.g., C:) by selecting it directly.



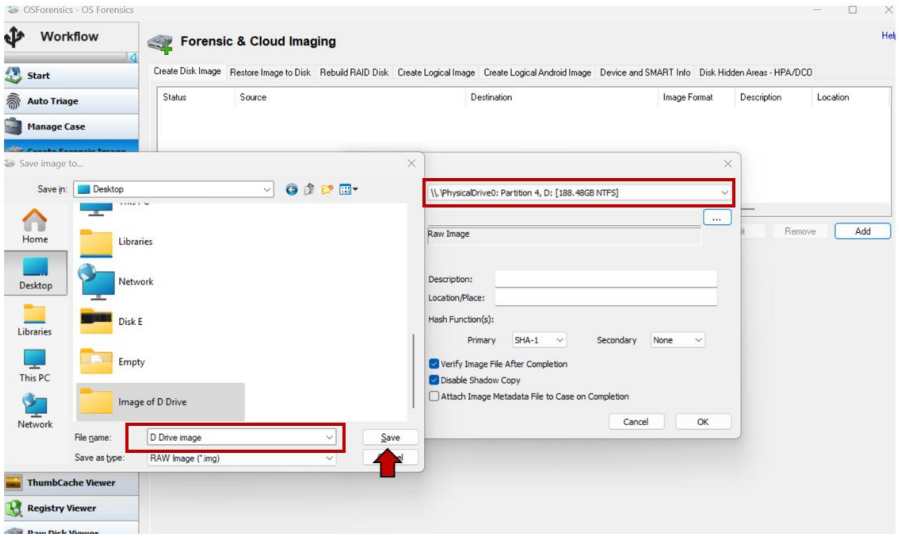


Creating a Forensic Image

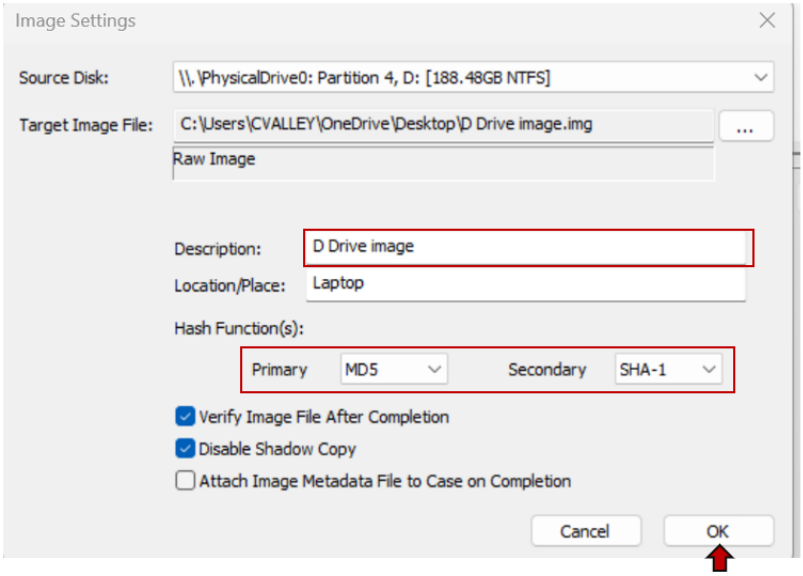
1. Click **Create Disk Image** from the left panel.
2. Click **Add** to begin setting up the imaging process.
3. In the **Image Settings** window:



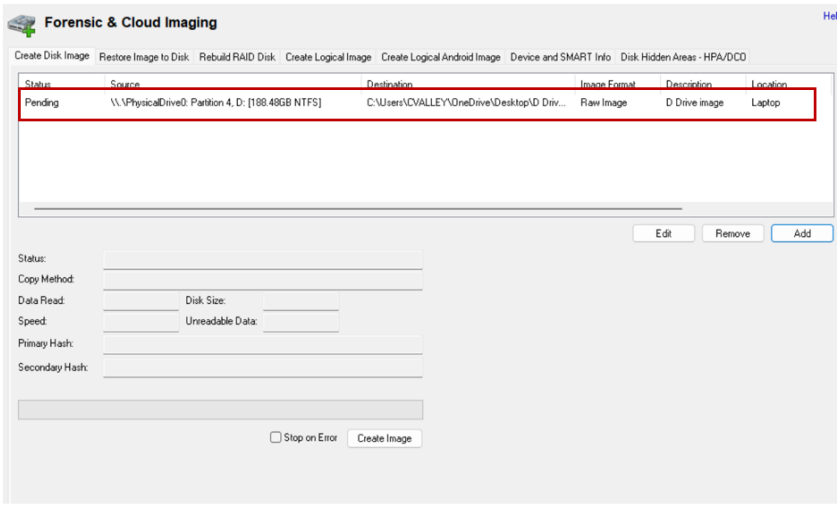
- Select the **Source Drive** you want to create an image of.
- Choose a destination folder under **Target Image File** where the image will be saved.



- Add a **Description** for the image (e.g., case reference, location).
- Select **Hash Functions** (e.g., MD5, SHA-1) for integrity verification.
- Click **OK** to begin image creation.

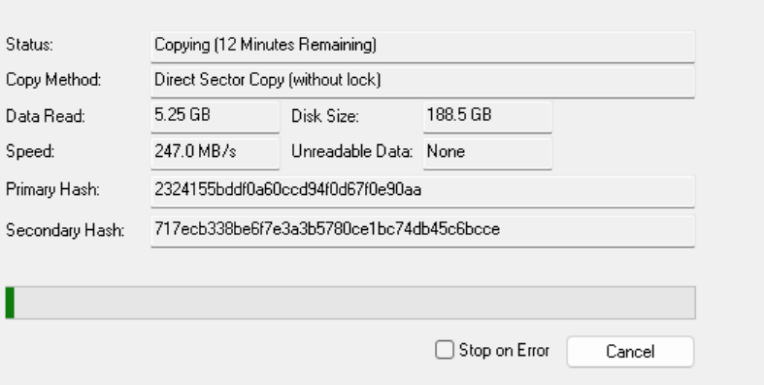


● A disk image of the selected drive will now be created, and you can later mount it for analysis.



"Forensics works on virtual created drive not on the actual drive to make analysis easy and the data remains safe"

Image of D Drive is in process…

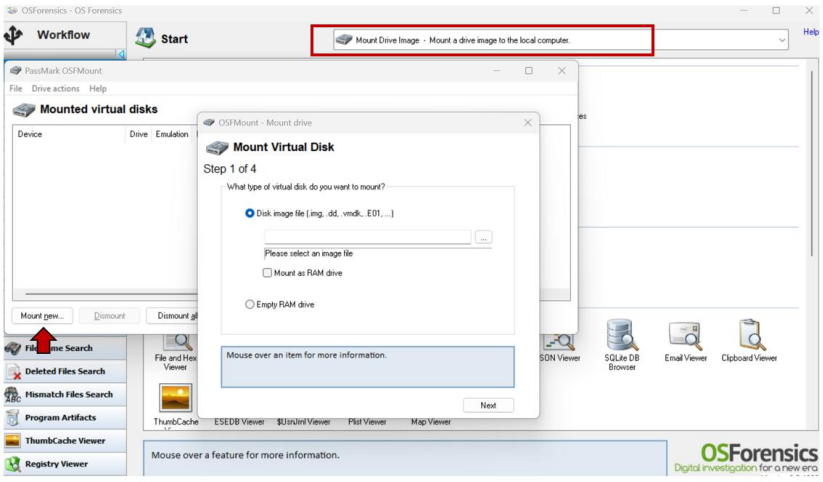


After Completion You will get a Disk image of the 188.5 Gb On the same Drive of the computer where you can perform analysis easily

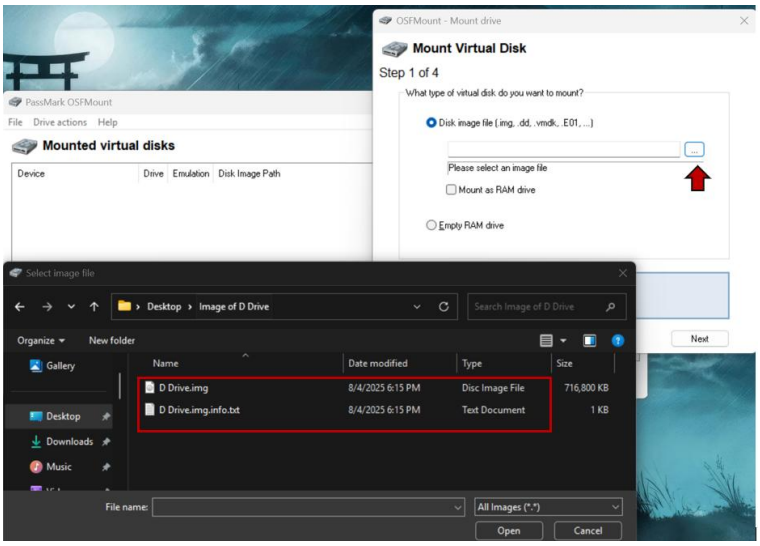
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Mounting a Forensic Image

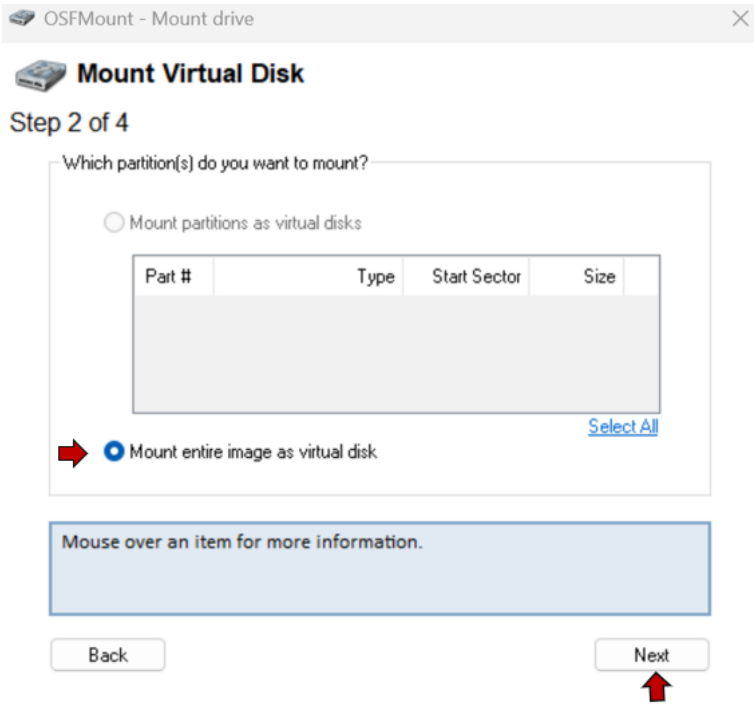
- Search **Mount Drive Image** from the search bar.
- Select **Mount new**



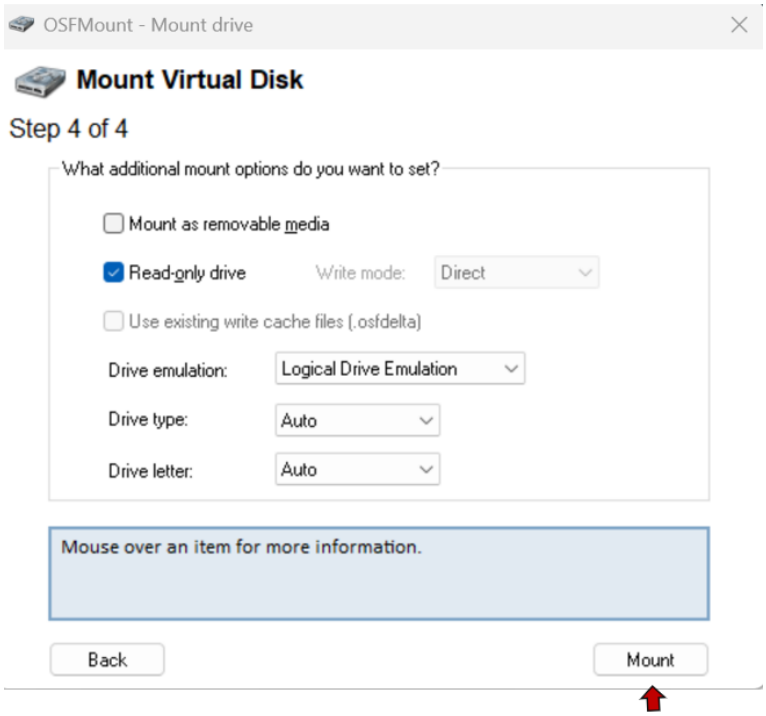
- Select the image file you previously created.



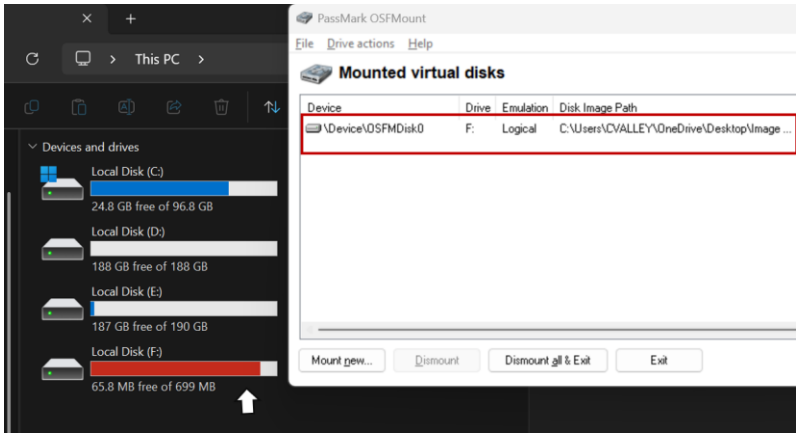
- Choose **Mount Entire Image** and click **Next**.



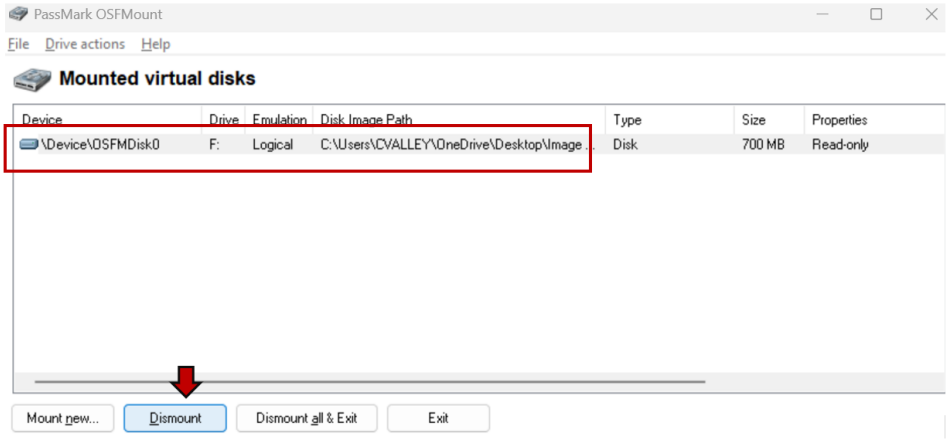
- Press **Mount** to complete the process.



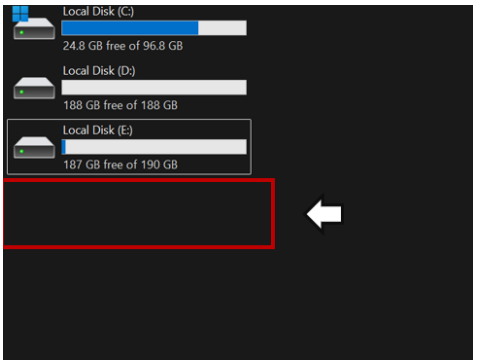
- Your image will appear as a virtual drive (e.g., F:\), ready for detailed analysis.



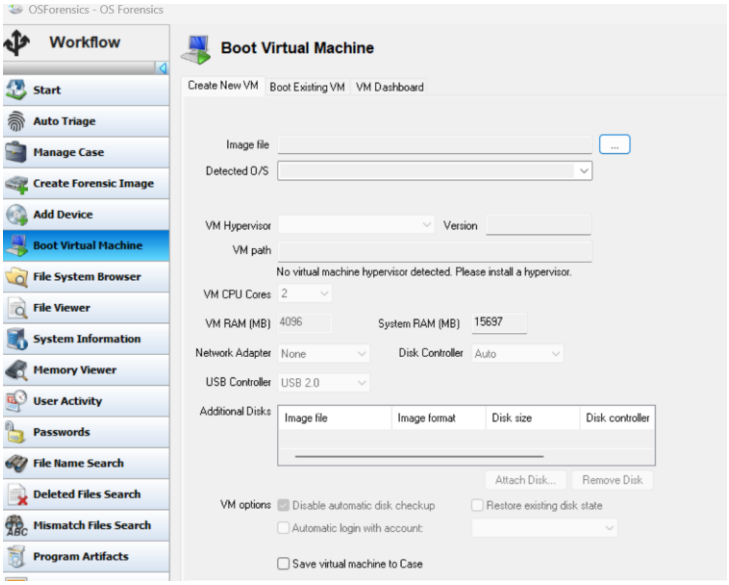
- To remove it, simply click on the device and choose **Dismount**.



- Drive has been Dismounted Successfully

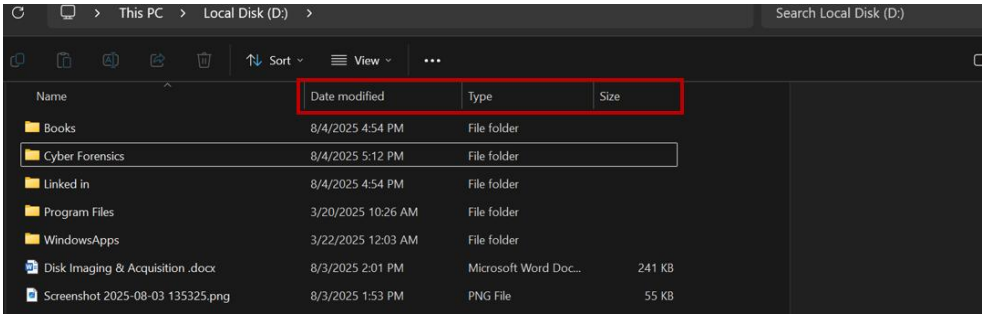


Menu for Mounting Through a virtual Machine

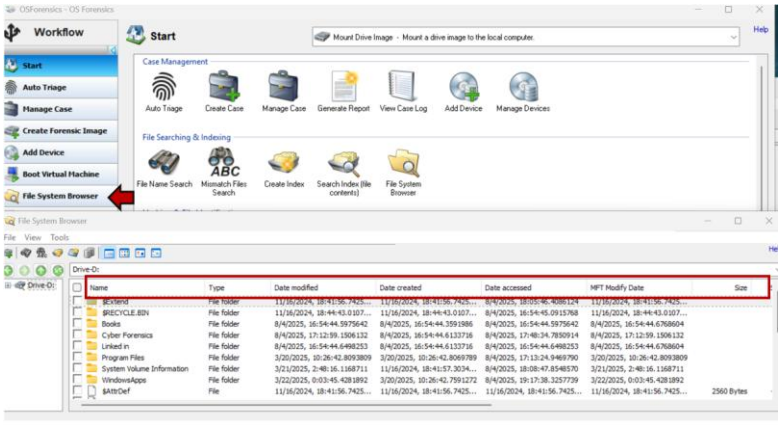


File System Browser

- A regular OS shows basic file info like size and type.

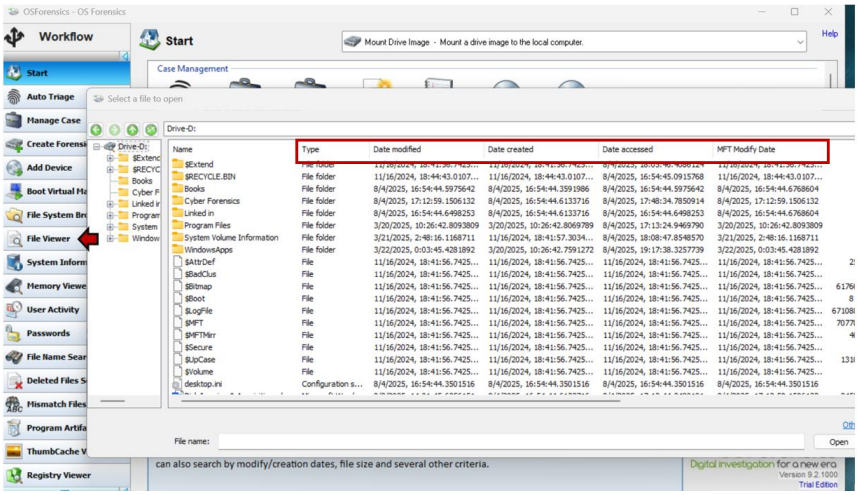


- OS Forensics provides **detailed metadata**, including file paths, creation time, access logs, etc.
- Click **File System Browser**, choose a drive, and view advanced file details.



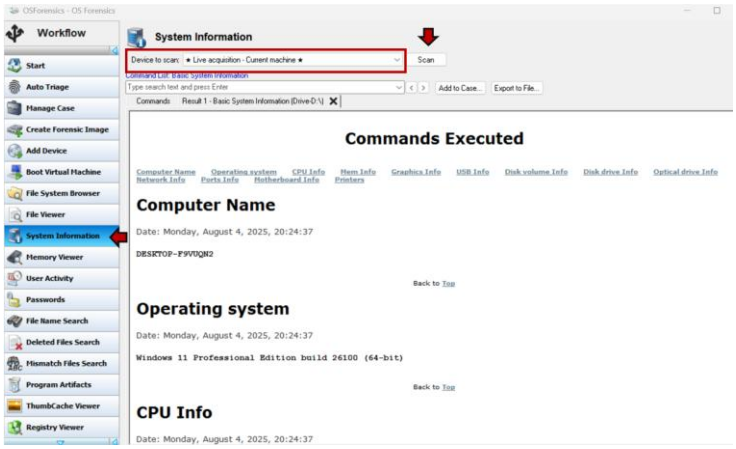
File Viewer

- Go to **File Viewer** to inspect detailed file information.
- Supports various file types including text, image, email, and binary.
- Useful for quickly examining suspicious files or document metadata.



System Information

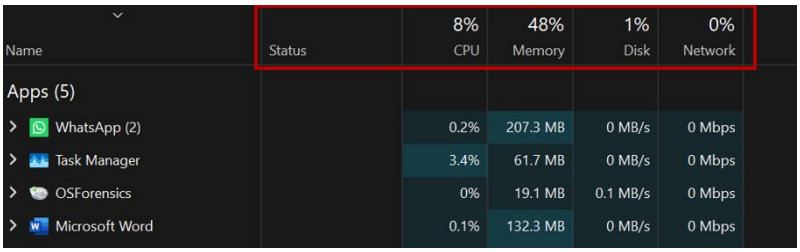
- Click **System Information** from the side panel.
- Select the device you want to scan.
- OS Forensics will display comprehensive details about the system, such as:
  - OS version
  - BIOS info
  - Installed hardware
  - Active network interfaces



Memory Viewer

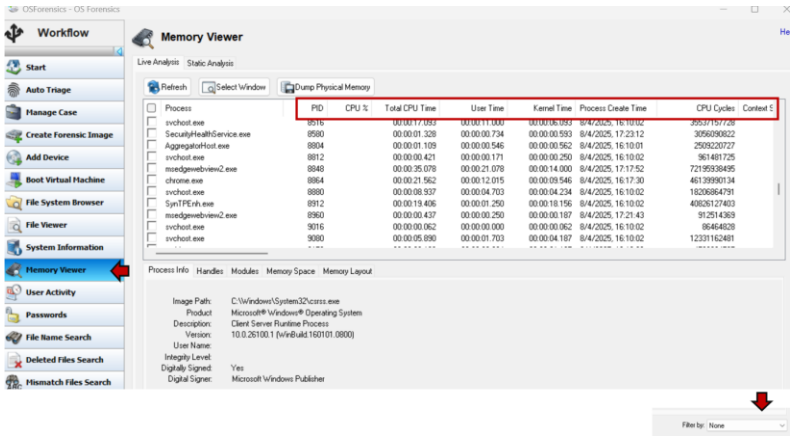
- Similar to Task Manager but with **more forensic-level detail**.
- Shows running processes, modules, paths, and memory regions.

task manager



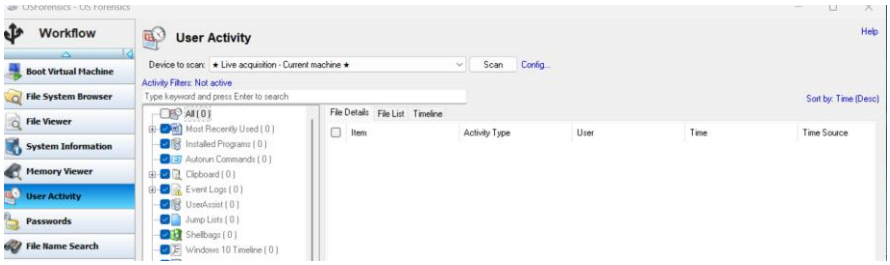
Using Memory viewer in OS Forensics

- Click **Memory Viewer** to analyze memory.
- You can filter results by clicking **Filter By** in the bottom-right corner.



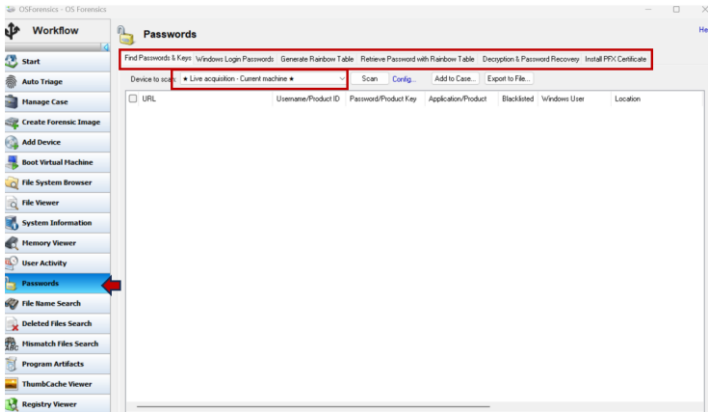
User Activity

- View recent files opened by the user.
- Check for suspicious remote access or unauthorized activity.
- OS Forensics logs actions like file openings, logins, and USB connections.



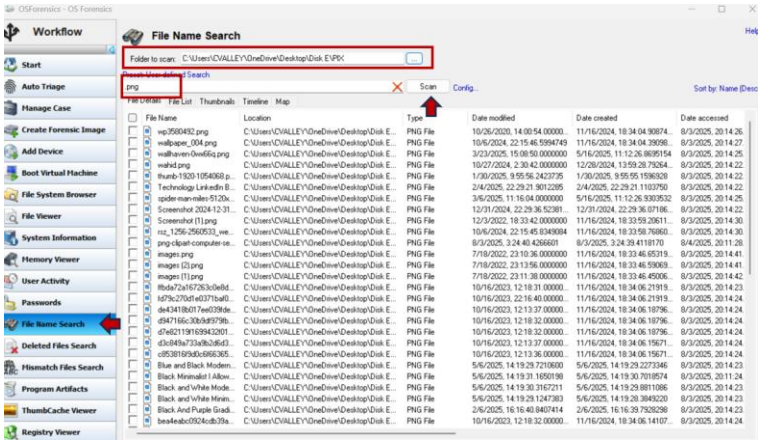
Password Recovery

- Use this feature to extract:
    - Saved browser passwords
    - Windows login credentials
    - System-stored credentials
- Only works on **live acquisitions** or mounted images that retain login data.



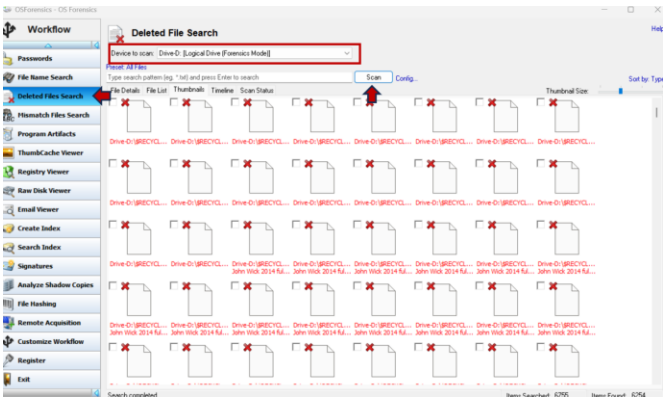
File Name Search

- Helps locate files by name or extension.
- Browse to a drive, enter keywords (e.g., .txt, .jpg), and run a search.
- Great for narrowing down specific file types across large volumes.



Deleted File Recovery

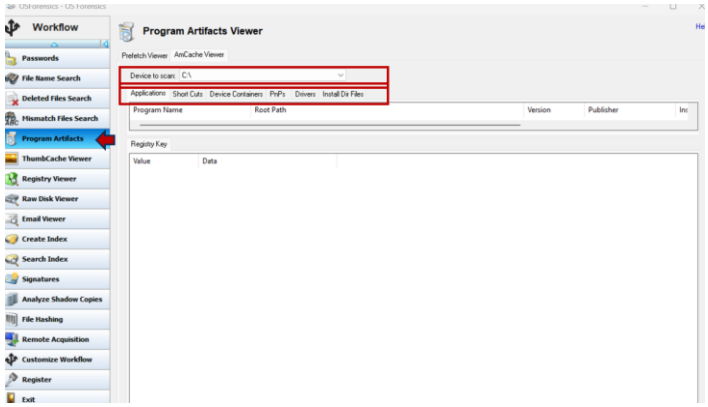
- Click **Deleted File Search**.
- Enter the drive path under **Device to Scan**.
- Begin scanning — OS Forensics will recover deleted files (even those emptied from Recycle Bin).



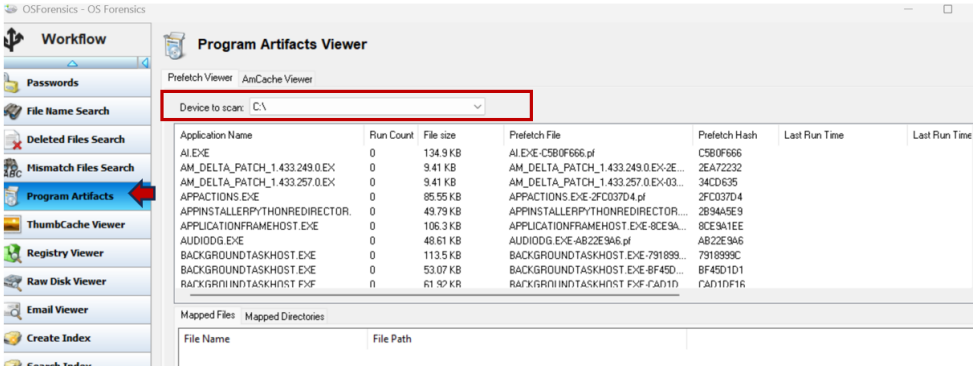
Program Artifacts & Am-Cache Viewer

- Click on **Program Artifacts**.
- Select a drive to analyze.

- **Useful for identifying:**
  - Installed programs
  - Application run history
  - Am-Cache data (executed binaries, install timestamps)

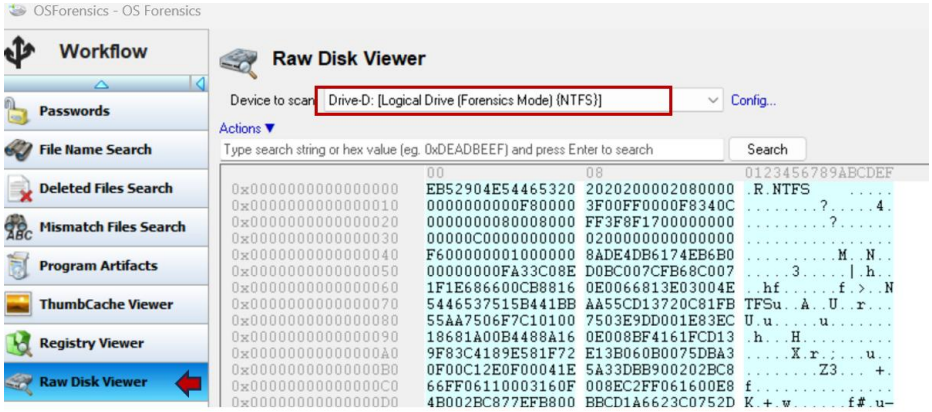


Helps in seeing the programs being running or software that has been installed on your system



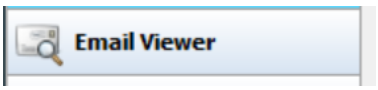
Raw Disk Viewer

- Shows low-level hexadecimal view of the disk.
- Ideal for analyzing raw data, deleted headers, or suspicious binary content.



Email Viewer

- Use when analyzing .PST or .OST files (from Outlook or similar programs).
- Allows you to browse and search through email content for forensic review.



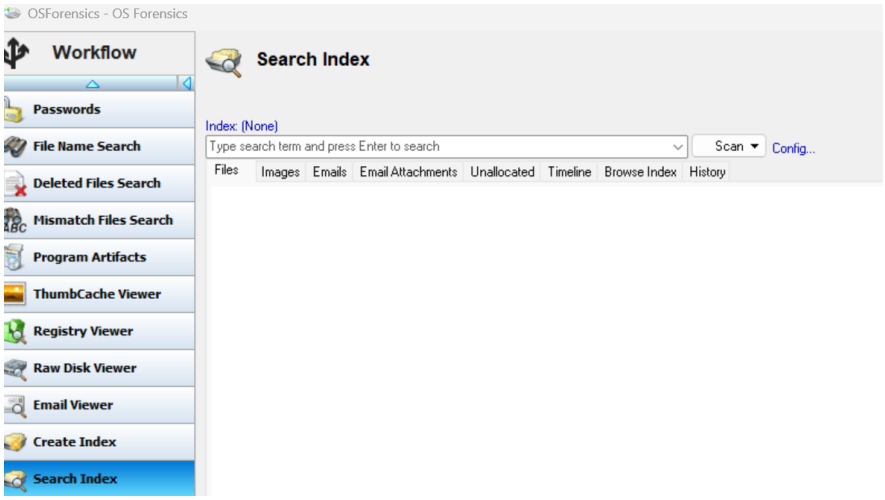
Index Creation

Click **Create Index** to generate a searchable database of all files.



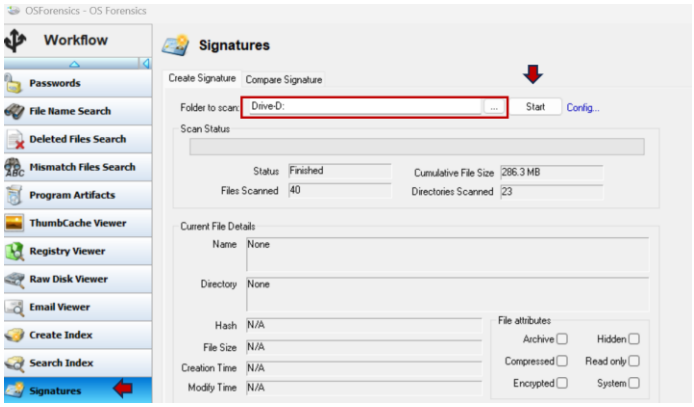
Index Search

Click **Search Index** to quickly find specific keywords, files, or metadata across the entire image.



Signature Comparison

- Click **Signature** → Select a drive → Click **Start** to generate a signature snapshot (e.g., Test01).
- Make changes (e.g., delete or add a file).

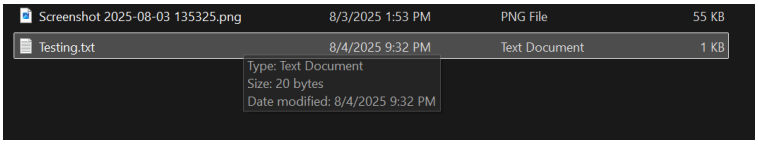




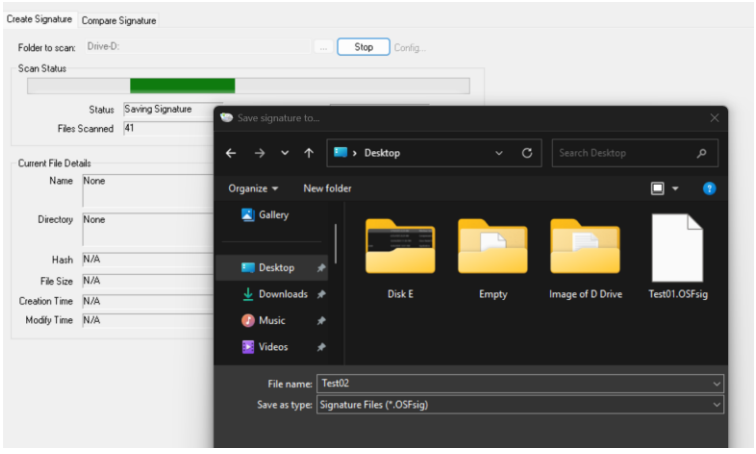
Signature 1 (Test01) is created.

I created 1 new file as a txt file and there is an old file which is screenshot of 3rd August 4, 2025

Del the old file and let remain the new txt file and see can OS Forensics will tell what change occur in our D Drive

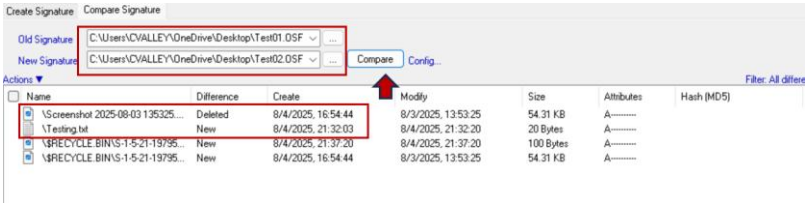


Performing Signature Create a second one for Test02.



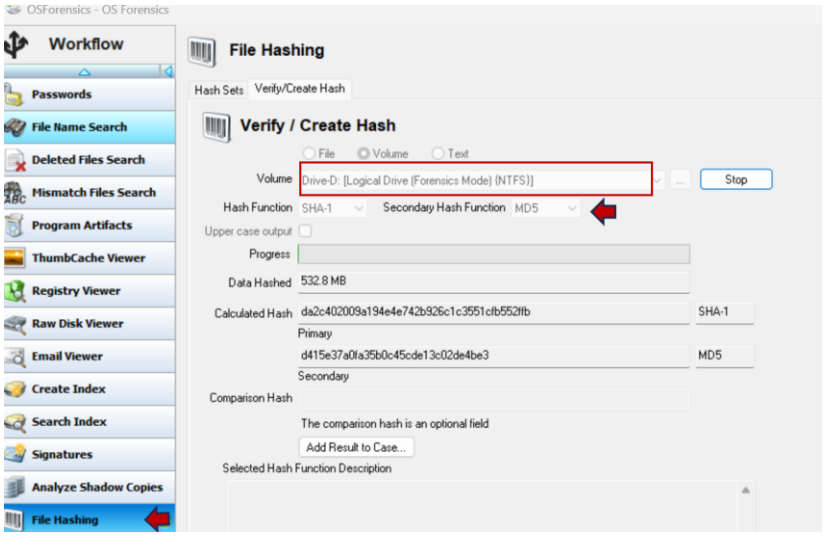
Click **Compare Signatures** → Select both snapshots → Run comparison.

- OS-Forensics will show exactly what files were added, removed, or changed — great for tampering detection.



File Hashing & Verification

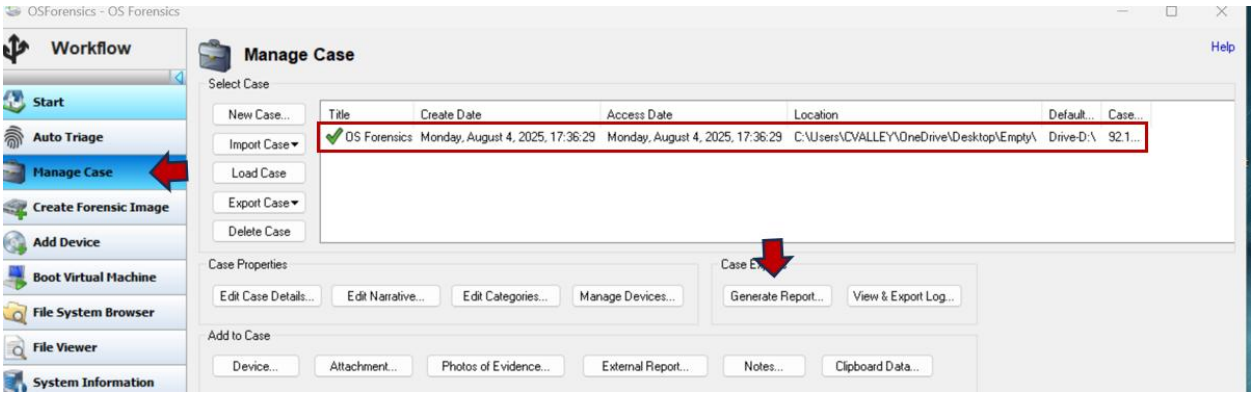
- Go to **File Hashing**.
- Select the file or volume to hash.
- Choose hash types (e.g., MD5, SHA-1).
- Run the hash function — the hash value will be generated.
- Re-run the hash later to detect unauthorized changes.
- You can add hash results to your case files for reporting and future validation.



Case Report Generation

After completing your analysis:

- Go to **Case Manager**.
- Add relevant findings and logs.
- Export a professional case report to summarize evidence and actions taken.



Conclusion

OS Forensics bridges multiple areas of digital forensic investigation under one platform — making it especially powerful for field agents, academic environments, and professional investigators. This guide demonstrates that OS Forensics can be used not just for disk imaging but also for live analysis, timeline reconstruction, hashing, and more. Its wide feature set makes it ideal for both beginners and professionals looking to expand their forensic toolkit.

